

Pair baseline: an unconditional normalized pair-correlation asymptotic

Research proof draft

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Task: proof. Assumptions: UNCONDITIONAL. This draft reconstructs [1, Lemmas 1–4, (2.2)–(2.16)], with locators referring to arXiv:2306.04799v1. It also reconstructs the zero-side comparison (2.18) and prime-side mean square (2.19), and assembles the normalized pair-correlation asymptotic (1.4) as Theorem 18, with status **proved-draft**. In the explicit-formula series, all nontrivial zeros $\rho = \beta + i\gamma = 1/2 + \delta_\rho + i\gamma$ are counted with multiplicity, including both signs of γ . The squared-norm identity instead uses the finite index set I_T of zero occurrences with $0 < \gamma \leq T$, again with multiplicity. Horizontal displacements remain explicit; the finite-height envelope uses a published theorem to set $\delta_\rho = 0$ only in its verified low range. Powers of positive real numbers use their real logarithms. Write $\Lambda(p^k) = \log p$ for a prime p and $k \geq 1$, and $\Lambda(n) = 0$ otherwise, including $n = 1$. For $X \geq 1$, $t \in \mathbb{R}$, put

$$\mathcal{S}(X, t) = \sum_{\rho} \frac{2X^{\delta_\rho + i(\gamma - t)}}{1 + ((t - \gamma) + i\delta_\rho)^2}, \quad (1)$$

$$\mathcal{P}(X, t) = \sum_{n \geq 1} \Lambda(n) n^{-1/2 - it} \min\{n/X, X/n\}. \quad (2)$$

The interpretation of these series is absolute convergence, proved below.

1 Analytic inputs and their provenance

The following published inputs are unconditional. The first four support the local Lemmas 1–4 reconstruction; the next two additionally support the finite-height envelope. The last two supply prime-distribution inputs for the prime-side mean square. They are imported facts, not claims of new proofs in this document. The ledger records their precise statements and source locators.

1. **ZETA-ANALYTIC-001:** the meromorphic continuation of zeta, its simple pole at 1 with residue 1, its simple trivial zeros at $-2, -4, \dots$, the location $0 < \Re \rho < 1$ of its other zeros, conjugation symmetry, and its absolutely convergent Euler product in $\Re s > 1$. We use

$$\zeta(s) = \chi(s) \zeta(1 - s), \quad \chi(s) = 2^s \pi^{s-1} \sin(\pi s/2) \Gamma(1 - s).$$

See [4, §§25.2(i), 25.2(iv), 25.4, 25.10(i)]; the displayed functional equation is (25.4.2), and the Euler product is (25.2.11). Differentiating the Euler product on compact subsets of $\Re s > 1$ gives

$$-\frac{\zeta'}{\zeta}(s) = \sum_{n \geq 1} \Lambda(n) n^{-s}. \quad (3)$$

The differentiation is justified by the majorant $\sum_{n \geq 2} (\log n) n^{-1-\varepsilon} < \infty$.

2. **ZETA-COUNT-001**: [2, Corollary 14.3, p. 454] gives $N_*(v) = \frac{v}{2\pi} \log \frac{v}{2\pi} - \frac{v}{2\pi} + O(\log v)$ for $v \geq 4$, with multiplicity and half-weight at $\gamma = v$. This is the unconditional corollary; the following Corollary 14.4 assumes RH and is not used. Lemma 1 below derives the inclusive endpoint and local counting versions needed here from this imported statement. The upstream proof of Riemann–von Mangoldt is not reconstructed.
3. **ZETA-CONTOUR-001**: [2, Lemma 12.2, p. 398] supplies heights $U_j \in [j, j+1]$, $j \geq 2$, avoiding zero ordinates, such that $\frac{\zeta'}{\zeta}(\sigma + iU_j) = O(\log^2 U_j)$, uniformly for $-1 \leq \sigma \leq 2$. Conjugation supplies the same bound at $-U_j$. Lemma 12.4, pp. 399–400, gives $\frac{\zeta'}{\zeta}(z) = O(\log(|z| + 1))$ uniformly for $\Re z \leq -1$ and $|z + 2m| \geq 1/4$ for every positive integer m . The cited proofs use no RH or simplicity hypothesis.
4. **GAMMA-DIGAMMA-001**: [2, Theorem C.1, (C.17), p. 523] gives $\psi(z) = \log z + O(1/|z|)$, uniformly in any fixed closed sector bounded away from the negative real axis and from zero; $\psi = \Gamma'/\Gamma$ and $\log z$ is principal. In particular fix an effective constant C_ψ such that

$$|\psi(z) - \log z| \leq C_\psi/|z| \quad (\Re z \geq 3/2). \quad (4)$$

The Euler–Maclaurin proof of that theorem yields an effective constant; we do not optimize or numerically evaluate it here.

5. **ZETA-KV-001**: Mossinghoff–Trudgian–Yang [3, Theorem 1.1, (1.5), p. 2] prove the following explicit Korobov–Vinogradov zero-free region:

$$\begin{aligned} \nu_{KV}(u) &= \frac{1}{(55241/1000)(\log u)^{2/3}(\log \log u)^{1/3}}, \\ \zeta(\sigma + it) \neq 0 \quad &\text{if} \quad |t| \geq 3, \quad \sigma \geq 1 - \nu_{KV}(|t|). \end{aligned} \quad (5)$$

Here $u \geq 3$, all logarithms are natural, the fractional powers are positive real powers, and $55241/1000 = 55.241$ exactly. Both boundary inequalities are inclusive. Therefore every nontrivial zero with $|\gamma| \geq 3$ satisfies the strict bound

$$\beta < 1 - \nu_{KV}(|\gamma|). \quad (6)$$

This applies to every occurrence, irrespective of multiplicity.

We checked the published version of record, including its stated domain and constants; we have not independently reconstructed its proof. Its Section 5 uses, among other inputs, the rigorous finite-height verification of Platt–Trudgian [5] up to $3 \cdot 10^{12}$; Sections 8–9 describe additional computational ingredients. The ledger records this external computational dependence with no locally replayed certificate. No unproved RH or simplicity assumption is imposed. Lemma 11 uses this input together with the following low-height theorem.

6. **ZETA-LOW-001**: Platt–Trudgian [5] prove that every nontrivial zero $\rho = \beta + i\gamma$ with

$$0 < \gamma \leq H, \quad H = 3 \cdot 10^{12}, \quad (7)$$

has $\beta = 1/2$. We import this assertion for every occurrence, without a simplicity assertion. The inclusive range is stated in the published abstract. We also consulted Theorem 1 and Section 2 of arXiv:2004.09765v1, a preprint version of the published work; that theorem gives the larger height 3,000,175,332,800. We use only the smaller published range (7). The proof uses rigorous Arb arithmetic and a variation of Turing’s method to account for all zeros in the range. Full comparison with the final journal text and independent computational replay remain pending. This finite-range published theorem is an unconditional input.

7. **PRIME-PNT-001**: [2, Theorem 6.9, (6.12)–(6.13), p. 179] supplies absolute effective constants $c, C > 0$ such that, for real $u \geq 2$,

$$|\psi(u) - u| + |\vartheta(u) - u| \leq Cu \exp(-c\sqrt{\log u}), \quad \psi(u) = \sum_{n \leq u} \Lambda(n), \quad \vartheta(u) = \sum_{p \leq u} \log p. \quad (8)$$

Endpoints have full weight. The classical zero-free-region proof on pp. 180–181 is unconditional and effective; its upstream analytic inputs are not reconstructed here.

8. **PRIME-SIEVE-001**: [2, Corollary 3.14, p. 97] implies that, for every $U \geq 4$ and positive even integer h ,

$$\#\{p \leq U : p, p+h \text{ prime}\} \ll g(h) \frac{U}{\log^2 U}, \quad g(h) = \prod_{\substack{p|h \\ p>2}} \frac{p-1}{p-2}. \quad (9)$$

The implied constant is absolute, effective, and independent of h . This is the $x=0, y=U$ specialization of the source, with its fixed twin-prime product and bounded $1+O(\log \log U / \log U)$ factor absorbed. The preceding elementary sieve proof supplies effectivity. This upper bound assumes no prime-pair asymptotic.

2 Inclusive counting and rational-kernel bounds

Lemma 1 (**PAIR-COUNT-001**; BGSTB Lemma 2). *Let $N(v)$ count all occurrences with $0 < \gamma \leq v$. Uniformly for $v \geq 3$,*

$$N(v) = \frac{v}{2\pi} \log \frac{v}{2\pi} - \frac{v}{2\pi} + O(\log v). \quad (10)$$

In particular $N(v) \sim v \log v / (2\pi)$ as $v \rightarrow \infty$. There is an absolute effective constant C_{count} such that

$$\#\{\rho : v \leq \gamma \leq v+1\} \leq C_{\text{count}} \log(|v|+2) \quad (v \in \mathbb{R}). \quad (11)$$

Endpoints and repeated occurrences have full weight in this inequality. Also $\#\{\rho : |\gamma| \leq V\} \ll V \log(V+2)$ for $V \geq 2$.

Proof. Write $f(v) = v \log(v/(2\pi)) / (2\pi) - v/(2\pi)$. For $v \geq 5$, monotonicity of the midpoint count gives

$$N_*(v-1) \leq N(v) \leq N_*(v+1).$$

Since $f'(v) = \log(v/(2\pi)) / (2\pi)$, both outer expressions are $f(v) + O(\log v)$ by the imported count. For $3 \leq v < 5$, use $N(v) \leq N_*(6)$ and boundedness of f on this interval. This proves (10) with an effective constant. For $v \geq 5$, the padded interval similarly gives

$$\#\{\rho : v \leq \gamma \leq v+1\} \leq N_*(v+2) - N_*(v-1) \ll \log(v+2).$$

The padding includes full multiplicities even at either endpoint.

For completeness, there are no nontrivial zeros with $\gamma = 0$. Indeed the paired series

$$H(s) = \sum_{n \geq 1} ((2n-1)^{-s} - (2n)^{-s})$$

converges locally uniformly for $\Re s > 0$: each difference equals $s \int_{2n-1}^{2n} u^{-s-1} du$, with a summable majorant on every compact subset. For $\Re s > 1$, absolute convergence gives $H(s) = (1 - 2^{1-s})\zeta(s)$; analytic continuation extends the identity to $\Re s > 0$, with the pole at 1 removed by the prefactor.

For real $0 < s < 1$, every paired term is positive and $1 - 2^{1-s} < 0$, so $\zeta(s) < 0$. Conjugation now gives $\#\{|\gamma| \leq V\} = 2N(V)$. For $v \leq -6$, conjugate the interval to $[-v-1, -v]$ and apply the positive-height bound. For $-6 < v < 5$, all its zeros are counted by $2N(8) \leq 2N_*(9)$, which has an effective bound. This proves (11) on the whole real axis. The total-count estimate follows from (10) for $V \geq 3$, and from $N(V) \leq N_*(4)$ for $2 \leq V < 3$. Dividing (10) by $v \log v / (2\pi)$ proves the asymptotic. $\square$

Lemma 2 (PAIR-COUNT-KERNEL-001). *Put $\kappa(y) = (1 + y^2)^{-1}$, and sum over all nontrivial zero occurrences. The following bounds have absolute effective constants:*

$$K(t) := \sum_{\rho} \kappa(t - \gamma) \ll \log(|t| + 2) \quad (t \in \mathbb{R}), \quad (12)$$

$$\sum_{|\gamma| > Z} \kappa(t - \gamma) \ll \frac{\log Z}{Z} \quad (T \geq 3, Z \geq 2T, 0 \leq t \leq T). \quad (13)$$

For $T \geq 3$, define the exterior and interior sums

$$B_T(t) = \sum_{\gamma \leq 0 \text{ or } \gamma > T} \kappa(t - \gamma), \quad A_T^{\text{ker}}(t) = \sum_{0 < \gamma \leq T} \kappa(t - \gamma).$$

Then

$$B_T(t) \ll \log(T + 2) \left(\frac{1}{1+t} + \frac{1}{1+T-t} \right) \quad (0 \leq t \leq T), \quad (14)$$

$$A_T^{\text{ker}}(t) \ll \frac{\log(T + 2)}{1 + \text{dist}(t, [0, T])} \quad (t \notin [0, T]). \quad (15)$$

In particular all the infinite sums displayed above converge absolutely.

Proof. We first record, uniformly for real $a \geq 0$, the elementary bounds

$$\sum_{n \geq 0} \frac{1}{1 + (a + n)^2} \ll \frac{1}{1 + a}, \quad \sum_{n \geq 0} \frac{\log(n + 2)}{1 + (a + n)^2} \ll \frac{\log(a + 2)}{1 + a}. \quad (16)$$

To prove these, set $D = 1 + a$. The $n \leq D$ terms have denominator at least $D^2/2$ and there are at most $2D$ of them. For $2^j D < n \leq 2^{j+1} D$, $j \geq 0$, there are at most $3 \cdot 2^j D$ terms, each with denominator at least $2^{2j} D^2$ and logarithmic numerator at most $\log D + (j+2) \log 2$. The convergent series $\sum 2^{-j}$ and $\sum (j+2) 2^{-j}$ give (16), with explicit effective constants if desired. All terms are nonnegative, so grouping needs only monotone convergence.

The distance shell $n \leq |t - \gamma| < n + 1$ is covered by two closed unit intervals. By (11) its count is at most $C \log(|t| + n + 3)$, even when shell endpoints are zero ordinates. Using $\log(|t| + n + 3) \leq \log(|t| + 2) + \log(n + 2)$ and (16) with $a = 0$ proves (12). For $|\gamma| > Z \geq 2T$, $|t - \gamma| \geq |\gamma|/2$. The total-count bound on the disjoint dyadic shells $2^j Z < |\gamma| \leq 2^{j+1} Z$ gives

$$\sum_{|\gamma| > Z} \kappa(t - \gamma) \ll \sum_{j \geq 0} \frac{\log(2^{j+1} Z + 2)}{2^j Z} \ll \frac{\log Z}{Z},$$

uniformly in the stated range, proving (13).

For $0 \leq t \leq T$, cover the lower exterior by $[-n-1, -n]$, $n \geq 0$. Its kernel sum is at most

$$C \sum_{n \geq 0} \frac{\log(n + 3)}{1 + (t + n)^2} \ll \frac{\log(t + 2)}{1 + t}.$$

Cover the upper exterior by $(T + n, T + n + 1]$; its sum is at most

$$C \sum_{n \geq 0} \frac{\log(T + n + 2)}{1 + (T - t + n)^2} \ll \frac{\log(T + 2) + \log(T - t + 2)}{1 + T - t}.$$

Here $\log(T + n + 2) \leq \log(T + 2) + \log(n + 2)$ and (16) justify the last inequality. These two estimates prove (14). For $t = -a < 0$, cover the interior zeros by $[n, n + 1) \cap (0, T]$; for $t = T + a > T$, use $(T - n - 1, T - n] \cap (0, T]$, in each case with $n \geq 0$. Every nonempty interval has at most $C \log(T + 2)$ occurrences and distance at least $a + n$ from t . The unweighted bound in (16) proves (15). Overlaps in closed covers only enlarge these nonnegative majorants; they do not alter any zero-sum convention. $\square$

3 Convergence and a Mellin kernel

Lemma 3 (PAIR-EF-CONVERGENCE-001). *The zero series (1) and prime series (2) converge absolutely and locally uniformly on $[1, \infty) \times \mathbb{R}$. Their sums are continuous there. For $U \geq 2(|t| + 2)$,*

$$\sum_{|\gamma| > U} \left| \frac{2X^{\delta_\rho + i(\gamma - t)}}{1 + ((t - \gamma) + i\delta_\rho)^2} \right| \ll X^{1/2} \frac{\log U}{U}, \quad (17)$$

with an absolute effective implicit constant.

Proof. For real v and $|\delta| < 1/2$,

$$|1 + (v + i\delta)^2| \geq \Re(1 + (v + i\delta)^2) = 1 + v^2 - \delta^2 \geq \frac{3}{4}(1 + v^2). \quad (18)$$

Thus each zero term is at most $(8/3)X^{1/2}/(1 + (t - \gamma)^2)$. Lemma 1 gives $\#\{\rho : |\gamma| \leq V\} \ll V \log(V + 2)$ for $V \geq 2$. If $|\gamma| > U \geq 2(|t| + 2)$, then $|t - \gamma| \geq |\gamma|/2$. Summing over $2^k U < |\gamma| \leq 2^{k+1}U$, $k \geq 0$, bounds the tail by

$$CX^{1/2} \sum_{k \geq 0} \frac{\log(2^{k+1}U + 2)}{2^k U} \ll X^{1/2} \frac{\log U}{U}.$$

For $1 \leq X \leq B$, $|t| \leq H$, this tail majorant is uniform; the remaining zero terms are finite and continuous. For primes, $\min\{n/X, X/n\} \leq B/n$ on the same compact set, so $B(\log n)n^{-3/2}$ is a summable majorant, independent of t . Every term is continuous even at $X = n$. The Weierstrass criterion and dominated convergence prove the assertions. In particular no principal-value ordering of the final zero series is required. $\square$

Lemma 4 (PAIR-EF-MELLIN-001). *For $y > 0$,*

$$k(y) := \frac{1}{2\pi i} \int_{2-i\infty}^{2+i\infty} \frac{2y^w}{w^2 - 1} dw = \begin{cases} y - y^{-1}, & y > 1, \\ 0, & 0 < y \leq 1. \end{cases} \quad (19)$$

The integral converges absolutely, including at $y = 1$.

Proof. On $\Re w = 2$ the integrand is $O_y((1 + |\Im w|^2)^{-1})$. For $y > 1$, move the line to $\Re w = -R$, $R \geq 2$, crossing residues y and $-y^{-1}$ at 1 and -1 . First send the horizontal heights to infinity with R fixed; each horizontal integral is $O_{y,R}(V^{-2})$. The new vertical integral is $O(y^{-R}/R)$, since $|w^2 - 1| \geq (R - 1)^2 + (\Im w)^2$. Then send $R \rightarrow \infty$. For $0 < y < 1$, moving right crosses no pole and the vertical bound is $O(y^R/R)$. At $y = 1$, moving right again gives $O(1/R)$; thus the value is zero. Equivalently the two left residues would cancel exactly. These steps prove the formula without an endpoint half-weight. $\square$

4 The exact identity and its endpoints

Proposition 5 (PAIR-EF-EXACT-001). *Let $s = 1/2 + it$. For every $X \geq 1$, $t \in \mathbb{R}$,*

$$\mathcal{S}(X, t) = -\mathcal{P}(X, t) - X^{-1} \frac{\zeta'}{\zeta}(s-1) + E_{\text{pole}}(X, t) + E_{\text{trivial}}(X, t), \quad (20)$$

where

$$E_{\text{pole}}(X, t) = \frac{2X^{1-s}}{s(2-s)} = \frac{2X^{1/2-it}}{(1/2+it)(3/2-it)}, \quad (21)$$

$$E_{\text{trivial}}(X, t) = 2 \sum_{m \geq 1} \frac{X^{-2m-s}}{(2m+s-1)(2m+s+1)}. \quad (22)$$

The trivial-zero series also converges absolutely and locally uniformly on $[1, \infty) \times \mathbb{R}$.

Proof. Right-line integral. Fix first $X > 1$, $t \in \mathbb{R}$, and define

$$K(z) = \frac{2X^{z-s}}{(z-s)^2-1}, \quad J = \frac{1}{2\pi i} \int_{5/2-i\infty}^{5/2+i\infty} -\frac{\zeta'}{\zeta}(z) K(z) dz.$$

There are no poles on this line. The Euler series gives a product majorant $2X^2 \sum_n \Lambda(n) n^{-5/2} / |(2+i(v-t))^2-1|$, integrable in v . Fubini therefore permits termwise integration. Set $w = z-s$; this is a shift of the vertical parameter by t , justified by absolute convergence. Lemma 4 yields

$$J = \sum_{n \geq 1} \Lambda(n) n^{-s} k(X/n) = \sum_{n \leq X} \Lambda(n) n^{-s} (X/n - n/X). \quad (23)$$

The last sum is finite, and its endpoint term is zero.

Contours and bounds. For an integer $M \geq 1$ take the left line $\Re z = -a$, $a = 2M+1$, and horizontal sides at $\pm U_j$ supplied by ZETA-CONTOUR-001. Choose j large enough that $U_j > 2(|t|+2)$. The rectangle is oriented with the right side upward. In $-1 \leq \Re z \leq 2$ the logarithmic derivative is $O(\log^2 U_j)$. On $2 \leq \Re z \leq 5/2$ its Euler series is bounded absolutely. On $-a \leq \Re z \leq -1$, Lemma 12.4 applies on these horizontal sides and gives $O_M(\log U_j)$. Since $|X^{z-s}| \leq X^2$ and the kernel denominator has size $\gg U_j^2$, both horizontal contributions satisfy

$$E_{\text{height}} = O_{X,t,M}(\log^2 U_j / U_j^2) \rightarrow 0. \quad (24)$$

This limit is taken with X, t, M fixed.

On the left line every trivial zero is at distance at least one. Lemma 12.4 bounds its logarithmic derivative by $C \log(a + |v| + 2)$, with C independent of M, v . Factoring the kernel denominator bounds it below by $(a - 1/2)^2 + (v - t)^2$. Hence the whole left-line integral is at most

$$\begin{aligned} |E_{\text{left}}| &\ll X^{-a-1/2} \int_{\mathbb{R}} \frac{\log(a + |v| + 2)}{(a - 1/2)^2 + (v - t)^2} dv \\ &\ll X^{-a-1/2} \frac{\log(a + |t| + 2)}{a}. \end{aligned} \quad (25)$$

For the second inequality use $v = t + (a - 1/2)y$, $a + |v| + 2 \leq (a + |t| + 2)(1 + |y|)$, and the finiteness of $\int_{\mathbb{R}} \log(1 + |y|)/(1 + y^2) dy$. After $j \rightarrow \infty$, send $M \rightarrow \infty$, with X, t fixed. The bound (25) tends to zero.

Residues. The residues of $-\frac{\zeta'}{\zeta}(z) K(z)$ are as follows; at a nontrivial zero its multiplicity multiplies the listed value.

Location	Residue
$z = \rho$	$2X^{\rho-s}/(1 - (\rho - s)^2)$
$z = 1$	$-E_{\text{pole}}(X, t)$
$z = -2m$	$-2X^{-2m-s}/((2m + s)^2 - 1)$
$z = s + 1$	$-X \frac{\zeta'}{\zeta}(s + 1)$
$z = s - 1$	$X^{-1} \frac{\zeta'}{\zeta}(s - 1)$

There are no coincidences: $\Re(s + 1) = 3/2$ and $\Re(s - 1) = -1/2$ avoid all zeta zeros and its pole, for every real t . Zeta is regular and nonzero at zero, so it supplies no additional residue. Also $1 - (\rho - s)^2 = 1 + ((t - \gamma) + i\delta_\rho)^2$. The nontrivial-zero residues converge absolutely by Lemma 3. For the trivial residues,

$$|(2m + s - 1)(2m + s + 1)| = \sqrt{((2m - 1/2)^2 + t^2)((2m + 3/2)^2 + t^2)} \geq m^2 + t^2. \quad (26)$$

Thus their absolute values are at most $2/m^2$, uniformly for $X \geq 1$ and real t . This proves the claimed local uniform convergence and justifies the residue limit as $M \rightarrow \infty$. The residue theorem now gives

$$J = \mathcal{S} - E_{\text{pole}} - E_{\text{trivial}} - X \frac{\zeta'}{\zeta}(s + 1) + X^{-1} \frac{\zeta'}{\zeta}(s - 1).$$

By (3) and (23),

$$J + X \frac{\zeta'}{\zeta}(s + 1) = -X^{-1} \sum_{n \leq X} \Lambda(n) n^{1-s} - X \sum_{n > X} \Lambda(n) n^{-1-s} = -\mathcal{P}(X, t).$$

This proves (20) for every $X > 1$, including prime powers.

Endpoint conventions and limit order. If $X = n = p^k$, the coefficient $X/n - n/X$ in J is zero. Any common endpoint convention $\theta \in [0, 1]$, including the Perron convention $\theta = 1/2$, multiplies this zero by θ . After combining with the full Euler series, the coefficient in $-\mathcal{P}$ is -1 , so the prime-power term is counted once. The min weight is continuous through every prime power, as are all terms of (20); the local majorants above justify continuity of the sums.

Finally let $X \downarrow 1$, with t fixed or in a fixed compact interval. The majorants of Lemma 3 and (26) justify dominated convergence. The remaining terms are continuous, since $s - 1 = -1/2 + it$ meets no zero or pole of zeta. This proves the identity at $X = 1$. The order used was: fixed $X > 1, t, M$; $U_j \rightarrow \infty$; then $M \rightarrow \infty$; then $X \downarrow 1$. No claim of uniformity of (24) in unbounded X, t is needed for this exact identity. $\square$

5 Uniform estimates: the reconstructed lemma

Lemma 6 (PAIR-EF-001; BGSTB Lemma 1). *Uniformly for $X \geq 1, t \in \mathbb{R}$,*

$$\mathcal{S}(X, t) = -\mathcal{P}(X, t) + X^{-1} \log(|t| + 2) + E_{\text{arch}}(X, t) + E_{\text{DS}}(X, t) + E_{\text{pole}}(X, t) + E_{\text{trivial}}(X, t), \quad (27)$$

$$\begin{aligned} |E_{\text{arch}}(X, t)| &\leq C_{\text{arch}} X^{-1}, & |E_{\text{DS}}(X, t)| &\leq C_{\text{DS}} X^{-1}, \\ |E_{\text{pole}}(X, t)| &\leq \frac{4X^{1/2}}{1 + t^2}, & |E_{\text{trivial}}(X, t)| &\leq \frac{12X^{-5/2}}{|t| + 2}, \end{aligned} \quad (28)$$

where the constants are effective and independent of X, t . Here $E_{\text{pole}}, E_{\text{trivial}}$ are the exact terms above and

$$\begin{aligned} E_{\text{arch}}(X, t) &= X^{-1} \left(-\frac{\chi'}{\chi}(-1/2 + it) - \log(|t| + 2) \right), \\ E_{\text{DS}}(X, t) &= X^{-1} \frac{\zeta'}{\zeta}(3/2 - it). \end{aligned} \quad (29)$$

In particular this is precisely the error structure of [1, Lemma 1, (2.2)] after combining the first two remainders.

Proof. Logarithmic differentiation of the functional equation on $w = -1/2 + it$, where all factors are finite and nonzero, gives

$$-\frac{\zeta'}{\zeta}(w) = -\frac{\chi'}{\chi}(w) + \frac{\zeta'}{\zeta}(1-w), \quad -\frac{\chi'}{\chi}(w) = -\log(2\pi) - \frac{\pi}{2} \cot(\pi w/2) + \psi(1-w).$$

No branch of $\log \zeta$ is needed for these logarithmic derivatives. On this line $|\cot(-\pi/4 + i\pi t/2)| = 1$, as follows by taking the moduli of sine and cosine. Put $z = 3/2 - it$, $r = |t|$. The principal argument has modulus at most $\pi/2$, and

$$\frac{1}{2} \leq \frac{\sqrt{r^2 + 9/4}}{r + 2} \leq 1.$$

Consequently $|\log z - \log(r + 2)| \leq \log 2 + \pi/2$. Using (4) proves the archimedean bound, for example with $C_{\text{arch}} = \log(2\pi) + \pi + \log 2 + 2C_\psi/3$. This argument includes $t = 0$ and bounded t , without an asymptotic substitution $\log |t|$ at zero.

Absolute convergence of the Euler series gives $C_{\text{DS}} = \sum_{n \geq 2} \Lambda(n) n^{-3/2} \leq \sum_{n \geq 2} (\log n) n^{-3/2} < \infty$. For example the latter can be bounded effectively by its first terms and an elementary integral; no prime number theorem is required. Since $\sqrt{(t^2 + 1/4)(t^2 + 9/4)} \geq (1 + t^2)/2$, (21) gives the stated pole bound. From (26),

$$|E_{\text{trivial}}(X, t)| \leq 2X^{-5/2} \sum_{m \geq 1} \frac{1}{m^2 + r^2}.$$

If $0 \leq r \leq 1$, the sum is at most $1 + \int_1^\infty v^{-2} dv = 2$, so twice the sum is at most $4 \leq 12/(r + 2)$. If $r \geq 1$, monotonicity gives $\sum_{m \geq 1} (m^2 + r^2)^{-1} \leq \int_0^\infty (v^2 + r^2)^{-1} dv = \pi/(2r)$. Then $\pi/r \leq 3\pi/(r + 2) \leq 12/(r + 2)$. These estimates prove (28); all constants are independent of the contour parameters, which have already been removed. Combining only now the two X^{-1} errors gives

$$\mathcal{S} = -\mathcal{P} + X^{-1}(\log(|t| + 2) + O(1)) + O\left(\frac{X^{1/2}}{1 + t^2}\right) + O\left(\frac{X^{-5/2}}{|t| + 2}\right).$$

□

6 The elementary integral for the squared norm

Lemma 7 (PAIR-NORM-INTEGRAL-001). *For $a \in \mathbb{C}$ with $|\Im a| < 1$, the following integral converges absolutely:*

$$J(a) = \int_{\mathbb{R}} \frac{du}{(1 + u^2)(1 + (u + a)^2)} = \frac{2\pi}{4 + a^2}. \quad (30)$$

In particular $J(0) = \pi/2$.

Proof. Put $f_a(u) = ((1+u^2)(1+(u+a)^2))^{-1}$. Its poles are $i, -i, -a+i, -a-i$, none on the real axis. For $|u| = R \geq 4(|a|+1)$,

$$|1+u^2| \geq R^2/2, \quad |1+(u+a)^2| \geq R^2/8, \quad |f_a(u)| \leq 16/R^4.$$

On the real line these bounds give an integrable tail; on compact intervals the integrand is continuous. Close $[-R, R]$ by the upper semicircle. Its integral has modulus at most $16\pi/R^3$, hence vanishes as $R \rightarrow \infty$ for fixed a . For $a \neq 0$, the two upper poles i and $-a+i$ are distinct, with residues

$$\frac{1}{2i(1+(a+i)^2)}, \quad \frac{1}{2i(1+(-a+i)^2)}.$$

The other two poles are in the lower half-plane because $|\Im a| < 1$. Multiplying the residue sum by $2\pi i$ and using

$$\frac{1}{a(a+2i)} + \frac{1}{a(a-2i)} = \frac{2}{a^2+4}$$

proves (30) for $a \neq 0$. The excluded values $a = \pm 2i$ in this algebra are outside the stated strip. For $a = 0$ the poles coalesce, so use instead the antiderivative

$$\int \frac{du}{(1+u^2)^2} = \frac{u}{2(1+u^2)} + \frac{1}{2} \arctan u.$$

Its limits give $J(0) = \pi/2$, the same value as the right side of (30). This handles the double poles directly. $\square$

7 The squared-norm identity with horizontal displacement

Define $N(T) = |I_T|$, $M_X = \max(X^{1/2}, X^{-1/2})$, and, for $X > 0$,

$$v_j(t) = \frac{X^{\delta_j + i\gamma_j}}{1 + ((t - \gamma_j) + i\delta_j)^2}, \quad \mathcal{V}_{X,T}(t) = \sum_{j \in I_T} v_j(t), \quad (31)$$

$$w(z) = \frac{4}{4 - z^2}, \quad \Phi(X, T) = \sum_{j,k \in I_T} X^{\rho_j - \rho_k} w(\rho_j - \rho_k). \quad (32)$$

Both definitions use all zero occurrences, with no diagonal exclusion. The set I_T is finite by the analytic structure of zeta on a compact rectangle; no quantitative zero-counting estimate is needed here.

Lemma 8 (PAIR-NORM-001; BGSTB Lemma 3). *For every $X > 0$ and $T \geq 3$,*

$$\Phi(X, T) = \frac{2}{\pi} \int_{\mathbb{R}} |\mathcal{V}_{X,T}(t)|^2 dt. \quad (33)$$

The integral is finite. For $R \geq 2(T+1)$ it satisfies the explicit tail estimate

$$\int_{|t| > R} |\mathcal{V}_{X,T}(t)|^2 dt \leq \frac{512}{27} \frac{M_X^2 N(T)^2}{R^3}. \quad (34)$$

Proof. Integrability and finite expansion. The elementary inequality (18) uses only $|\delta_j| < 1/2$ and therefore also applies for $0 < X < 1$. Since $|X^{\delta_j + i\gamma_j}| \leq M_X$,

$$|v_j(t)| \leq \frac{4M_X}{3(1 + (t - \gamma_j)^2)}.$$

Consequently each v_j lies in $L^2(\mathbb{R})$; indeed $\int |v_j|^2 \leq (16/9)M_X^2 J(0) = 8\pi M_X^2/9$. Each $v_j \overline{v_k}$ is integrable by Cauchy–Schwarz. The sum has finitely many terms, so expanding its squared modulus and integrating term by term is valid by linearity for integrable functions. For $|t| > R \geq 2(T+1)$, $|t - \gamma_j| \geq |t|/2$, whence

$$|\mathcal{V}_{X,T}(t)|^2 \leq \frac{256M_X^2 N(T)^2}{9t^4}.$$

Integrating over both tails gives (34). This bound displays the dependence on $X, T, N(T)$; no bound uniform as $X \rightarrow 0$, $X \rightarrow \infty$, or $T \rightarrow \infty$ is claimed.

The integral for one ordered pair. For fixed j, k , set

$$d = \gamma_j - \gamma_k, \quad h = \delta_j + \delta_k, \quad a = d - ih = -i(\rho_j + \overline{\rho_k} - 1).$$

Then $|\Im a| = |h| < 1$. In the integral of $v_j \overline{v_k}$, substitute $u = t - \gamma_j + i\delta_j$. This takes the real line to the line $\Im u = \delta_j$, and gives

$$\int_{\mathbb{R}} v_j(t) \overline{v_k(t)} dt = X^{h+id} \int_{\mathbb{R}+i\delta_j} f_a(u) du. \quad (35)$$

Both line integrals are oriented toward increasing real part. The absolute convergence already established justifies translating the real parameter and choosing symmetric real endpoints in u .

The contour shift. The imaginary parts of the four poles of f_a are $1, -1, h+1, h-1$. Let $b_- = \min(0, \delta_j)$ and $b_+ = \max(0, \delta_j)$. The first two poles lie outside $[b_-, b_+]$ because $|\delta_j| < 1/2$. For the other two,

$$h+1 = \delta_j + (\delta_k + 1) > \delta_j + 1/2 > b_+, \quad h-1 = \delta_j + (\delta_k - 1) < \delta_j - 1/2 < b_-.$$

Thus no pole lies in the closed strip between the two lines, including when δ_j is negative or when poles coincide. For $u = \pm R + iy$, $b_- \leq y \leq b_+$, and $R \geq 4(|a| + 1)$, the modulus bounds used in Lemma 7 again give $|f_a(u)| \leq 16/R^4$: use $|u| \geq R$ and $|u + a| \geq R - |a|$. The total length of the two joining segments is $2|\delta_j| < 1$, so their total contribution has modulus at most $16/R^4$. Cauchy's theorem on this rectangle, followed by $R \rightarrow \infty$ with the pair fixed, therefore gives

$$\int_{\mathbb{R}+i\delta_j} f_a(u) du = \int_{\mathbb{R}} f_a(u) du = \frac{2\pi}{4 - (\rho_j + \overline{\rho_k} - 1)^2}. \quad (36)$$

If $\delta_j = 0$ there is no shift. If $a = 0$ use the double-pole case of Lemma 7. This case includes distinct reflected off-line zeros with the same ordinate; it is not restricted to equality of complex zeros.

Normalization and reflection. Combining (35)–(36) and the finite expansion yields

$$\frac{2}{\pi} \int_{\mathbb{R}} |\mathcal{V}_{X,T}(t)|^2 dt = \sum_{j,k \in I_T} X^{\rho_j + \overline{\rho_k} - 1} \frac{4}{4 - (\rho_j + \overline{\rho_k} - 1)^2}. \quad (37)$$

The functional equation and conjugation imply that $\rho_k \mapsto 1 - \overline{\rho_k}$ preserves the positive ordinate and multiplicity: the factor χ is holomorphic and nonzero in $0 < \Re s < 1$. It therefore induces a permutation of the finite set of occurrences I_T , including occurrences at $\gamma = T$. Reindexing k in the entire sum turns $\rho_j + \overline{\rho_k} - 1$ into $\rho_j - \rho_k$, proving (33). This proof holds with X, T fixed; its only limits are the elementary integration contours tending to infinity. There is no limit in zero height and no infinite zero-sum interchange. $\square$

The reflected sum in (37) and the original sum (32) generally have different index diagonal contributions. A reflected-sum term with $j = k$ is $X^{2\delta_j}/(1 - \delta_j^2)$, whereas an original index-diagonal term is 1. The permutation equates the complete sums, not these subsets. Repeated occurrences remain separate indices throughout.

Corollary 9 (PAIR-POSITIVITY-001). *For $X > 0, T \geq 3$, $\Phi(X, T)$ is real and nonnegative, and $\Phi(X^{-1}, T) = \Phi(X, T)$. Consequently $\mathcal{F}_T(\alpha) = \Phi(T^\alpha, T)/(T \log T/(2\pi))$ is real, nonnegative and even for every real α .*

Proof. Reality and nonnegativity follow from (33). For the inversion identity, exchange j, k in the finite sum (32) and use $w(-z) = w(z)$. The normalization is positive for $T \geq 3$, and $T^{-\alpha} = (T^\alpha)^{-1}$. No asymptotic formula is used. $\square$

8 Height truncation with separate errors

For $X \geq 1$, define

$$\mathcal{U}_X(t) = \sum_{\rho} v_{\rho}(t), \quad \mathcal{U}_{X,Z}(t) = \sum_{|\gamma| \leq Z} v_{\rho}(t), \quad L(X, T) = 4 \int_0^T |\mathcal{U}_X(t)|^2 dt, \quad (38)$$

where $v_{\rho}(t)$ has the definition (31). The full sum includes all nontrivial zero occurrences, and the truncated sum retains full multiplicity at both $\gamma = Z$ and $\gamma = -Z$. Lemma 3 gives absolute and local uniform convergence. More precisely $\mathcal{S}(X, t) = 2X^{-it}\mathcal{U}_X(t)$, so

$$L(X, T) = \int_0^T |\mathcal{S}(X, t)|^2 dt. \quad (39)$$

This identifies the integral with Lemma 1 without changing a normalization. Define the finite-height horizontal bound

$$B(Z) = \max(\{0\} \cup \{\delta_{\rho} : |\gamma| \leq Z\}), \quad \eta_*(Z) = \frac{1}{2} - B(Z). \quad (40)$$

The zero set in this maximum is finite, so $0 \leq B(Z) < 1/2$ and $0 < \eta_*(Z) \leq 1/2$, including an empty zero set. The definition uses actual zeros, not a zero-density or zero-free-region hypothesis. The definition alone supplies no quantitative lower bound for $\eta_*(Z)$ as $Z \rightarrow \infty$. For every retained occurrence and $X \geq 1$,

$$|v_{\rho}(t)| \leq \frac{4}{3} X^{B(Z)} \kappa(t - \gamma). \quad (41)$$

Lemma 10 (PAIR-TRUNC-001; BGSTB Lemma 4). *For $X \geq 1, T \geq 3, Z \geq 2T$, one has the exact decomposition*

$$L(X, T) = 2\pi\Phi(X, T) + E_{\text{trunc}} + E_{\text{height}} + E_{\text{extension}}, \quad (42)$$

where

$$E_{\text{trunc}} = 4 \int_0^T (|\mathcal{U}_X(t)|^2 - |\mathcal{U}_{X,Z}(t)|^2) dt, \quad (43)$$

$$E_{\text{height}} = 4 \int_0^T (|\mathcal{U}_{X,Z}(t)|^2 - |\mathcal{V}_{X,T}(t)|^2) dt, \quad (44)$$

$$E_{\text{extension}} = -4 \int_{\mathbb{R} \setminus [0, T]} |\mathcal{V}_{X,T}(t)|^2 dt. \quad (45)$$

Uniformly throughout this range, with absolute effective constants,

$$|E_{\text{trunc}}| \ll XT \left(\frac{\log(T+2) \log Z}{Z} + \frac{\log^2 Z}{Z^2} \right) \ll \frac{XT \log^2 Z}{Z}, \quad (46)$$

$$|E_{\text{height}}| \ll X^{2B(Z)} \log^3 T, \quad (47)$$

$$|E_{\text{extension}}| \ll X^{2B(Z)} \log^2 T, \quad E_{\text{extension}} \leq 0. \quad (48)$$

Consequently, for all $X \geq 1, T \geq 3$, with $Z_* = T \log^2 T$,

$$\Phi(X, T) = \frac{L(X, T)}{2\pi} + O(X^{2B(Z_*)} \log^3 T) + O(X). \quad (49)$$

The first error can equivalently be written using $\eta_*(Z_*)$ as $O(X^{1-2\eta_*(Z_*)} \log^3 T)$. More generally, it has that bound with any $0 \leq \eta(Z_*) \leq 1/2$ for which the established envelope $B(Z_*) \leq 1/2 - \eta(Z_*)$ holds. The implicit constants do not depend on the choice of this envelope.

Proof. Removing the distant zeros. By (18), (12) and (13), for $0 \leq t \leq T$,

$$|\mathcal{U}_X(t)| \ll X^{1/2} \log(T+2), \quad |R_{X,Z}(t)| \ll X^{1/2} \frac{\log Z}{Z}, \quad R_{X,Z} = \mathcal{U}_X - \mathcal{U}_{X,Z}.$$

These are bounds for the absolutely convergent sums, including their absolute-value majorants. In particular L is finite. Using the identity

$$|A|^2 - |A - R|^2 = 2\Re(A\overline{R}) - |R|^2$$

with $A = \mathcal{U}_X(t)$, integration yields the first bound in (46). Its quadratic term is retained. Since $T+2 \leq Z$ for $T \geq 3, Z \geq 2T$ and $Z \geq 6$, the simpler second bound follows. No infinite double sum is expanded.

Removing the lower and upper exterior occurrences. Every occurrence of I_T is included in $|\gamma| \leq Z$. By (41), both $|\mathcal{U}_{X,Z}(t)|$ and $|\mathcal{V}_{X,T}(t)|$ are at most $(4/3)X^{B(Z)}K(t)$, and

$$|\mathcal{U}_{X,Z}(t) - \mathcal{V}_{X,T}(t)| \leq \frac{4}{3}X^{B(Z)}B_T(t).$$

Dropping the cutoff Z enlarges only these nonnegative kernel majorants; we do not apply $X^{B(Z)}$ to unretained zero weights. For complex A, C , $||A|^2 - |C|^2| \leq |A - C|(|A| + |C|)$. Thus (12) and (14) give

$$\begin{aligned} |E_{\text{height}}| &\ll X^{2B(Z)} \int_0^T K(t) B_T(t) dt \\ &\ll X^{2B(Z)} \log^2(T+2) \int_0^T \left(\frac{1}{1+t} + \frac{1}{1+T-t} \right) dt \\ &\ll X^{2B(Z)} \log^2(T+2) \log(1+T). \end{aligned}$$

This proves (47), because $T \geq 3$. In particular no lower bound on the distance from T to a zero ordinate is needed. All occurrences at $\gamma = T$ remain inside.

Extending the integral. The same horizontal bound, now applied only to I_T , and (15) give

$$|E_{\text{extension}}| \ll X^{2B(Z)} \log^2(T+2) 2 \int_0^\infty \frac{da}{(1+a)^2} \ll X^{2B(Z)} \log^2 T.$$

The sign follows from its definition. Telescoping the three error definitions, and then using (33), gives

$$L = 4 \int_{\mathbb{R}} |\mathcal{V}_{X,T}(t)|^2 dt + E_{\text{trunc}} + E_{\text{height}} + E_{\text{extension}} = 2\pi\Phi + E_{\text{trunc}} + E_{\text{height}} + E_{\text{extension}}.$$

This establishes (42). All integrals are absolutely convergent by the displayed majorants. The sole infinite zero sum has the locally uniform interpretation already proved in Lemma 3; every subsequent difference and finite sum is at fixed X, T, Z .

Choosing the auxiliary height and treating small T . For $T \geq 5$, $Z_* = T \log^2 T \geq 2T$, and $\log Z_* = \log T + 2 \log \log T \ll \log T$. Hence $E_{\text{trunc}} = O(X)$; the other two errors give (49). The choice $Z_* \geq 2T$ is not valid for every $T \geq 3$, so we treat $3 \leq T < 5$ separately. The global kernel bound gives $L(X, T) \ll XT \log^2(T+2) \ll X$ on that bounded range. For $z = \rho_j - \rho_k = b + id$, $|b| < 1$ gives

$$|4 - z^2| \geq 4 - b^2 + d^2 \geq 3 + d^2, \quad |X^z| \leq X.$$

Consequently $|\Phi(X, T)| \leq (4/3)XN(5)^2 \ll X$, with an effective constant from Lemma 1. Thus $\Phi - L/(2\pi) = O(X)$ uniformly for $3 \leq T < 5$; the first error in (49) can be taken as zero there. This uses no numerical knowledge of the first zeta zero. Finally $2B(Z_*) = 1 - 2\eta_*(Z_*)$, and any other established envelope gives the claimed weaker error since $X \geq 1$. No quantitative zero-free region, RH, or simplicity enters the proof. $\square$

9 An explicit finite-height horizontal envelope

Lemma 11 (PAIR-ENVELOPE-001). *For every real $Z \geq 3$, the definitions (5) and (40) satisfy*

$$0 < \nu_{\text{KV}}(Z) < \frac{1}{2}, \quad B(Z) < \frac{1}{2} - \nu_{\text{KV}}(Z), \quad \eta_*(Z) > \nu_{\text{KV}}(Z). \quad (50)$$

Every nontrivial zero occurrence with $|\gamma| \leq Z$ satisfies

$$|\delta_\rho| \leq B(Z) < \frac{1}{2} - \nu_{\text{KV}}(Z). \quad (51)$$

Both height endpoints are included with full multiplicity. The claims about B and η_ also hold when the finite zero set is empty. Thus $\eta(Z) = \nu_{\text{KV}}(Z)$, for $Z \geq 3$, is an explicit admissible envelope in Lemma 10.*

Proof. Size and monotonicity. First, elementary integrals give

$$\log 3 = 2 \int_0^{1/2} \frac{dt}{1-t^2} > 2 \int_0^{1/2} (1+t^2) dt = \frac{13}{12}, \quad \log \log 3 > \log(13/12) = \int_1^{13/12} \frac{dt}{t} > \frac{1}{13}.$$

The first strict inequality follows from $(1-t^2)^{-1} - (1+t^2) = t^4/(1-t^2) > 0$ for $0 < t \leq 1/2$. Since $55241/1000 > 5$, these inequalities imply

$$\nu_{\text{KV}}(3)^{-3} = (55241/1000)^3 (\log 3)^2 \log \log 3 > \frac{125}{13} > 8.$$

Consequently $0 < \nu_{\text{KV}}(3) < 1/2$. For $u \geq 3$ all logarithms occurring in the definition are positive, and logarithmic differentiation gives

$$\frac{\nu'_{\text{KV}}(u)}{\nu_{\text{KV}}(u)} = -\frac{2}{3u \log u} - \frac{1}{3u \log u \log u} < 0. \quad (52)$$

Thus ν_{KV} is positive and decreasing on $[3, \infty)$; in particular $q_Z := 1/2 - \nu_{KV}(Z) > 0$.

All retained heights. Fix $Z \geq 3$. Lemma 1 excludes nontrivial real zeros, so $\gamma = 0$ contributes no occurrence. For $0 < |\gamma| < 3$, conjugation reduces the assertion to the positive range in (7), and hence $\delta_\rho = 0 < q_Z$. For $3 \leq |\gamma| \leq Z$, (6) and (52) give

$$\delta_\rho < \frac{1}{2} - \nu_{KV}(|\gamma|) \leq \frac{1}{2} - \nu_{KV}(Z) = q_Z.$$

In particular $|\gamma| = 3$ uses the inclusive KV threshold, and $|\gamma| = Z$ is allowed even when the second inequality is equality.

The maximum defining $B(Z)$ is over a finite collection by Lemma 1. Every entry, including the auxiliary 0, is strictly below q_Z ; hence $B(Z) < q_Z$. If there are no retained zeros, this says $B(Z) = 0 < q_Z$. Subtracting from $1/2$ gives the asserted bound for η_* . Reflection $\rho \mapsto 1 - \bar{\rho}$ preserves the ordinate and multiplicity and changes δ_ρ to $-\delta_\rho$. Both displacements occur in the maximum, so $|\delta_\rho| \leq B(Z)$, proving (51). Repeated occurrences do not change a maximum.

Everything is at fixed finite Z ; no sum, integral or limit is interchanged. The bounds hold for every $Z \geq 3$, with the displayed effective constant. Since $Z_* = T \log^2 T \geq 3$ for $T \geq 3$, this choice of η is admissible throughout the final range of Lemma 10. $\square$

10 Uniform comparison on the pair-theorem range

Lemma 12 (PAIR-COMPARE-001). *Uniformly for $T \geq 3$ and $1 \leq X \leq T$,*

$$L(X, T) = 2\pi\Phi(X, T) + O(T) + O(X), \quad (53)$$

with absolute effective implied constants. Both endpoints in X are included, and the zero sums retain the endpoint and multiplicity conventions of Lemma 10. More explicitly, for $Z_ = T \log^2 T$,*

$$X^{1-2\nu_{KV}(Z_*)} \log^3 T \leq C_* T, \quad C_* = 18! 84^{18}. \quad (54)$$

The constant C_ bounds this auxiliary expression; the constants in (53) also include those of Lemma 10.*

Proof. Put $y = \log T > 1$, $z = \log Z_* = y + 2 \log y > 1$, and $\nu = \nu_{KV}(Z_*)$. Lemma 11 gives $0 < \nu < 1/2$. Thus $X \mapsto X^{1-2\nu}$ is increasing and

$$X^{1-2\nu} \log^3 T \leq T^{1-2\nu} y^3 = T y^3 e^{-2\nu y} \quad (1 \leq X \leq T). \quad (55)$$

We bound the last factor for every $T \geq 3$. Since $\log y \leq y$, one has $z \leq 3y$. For $v \geq 1$, the derivative of $\sqrt{v} - \log v$ is $(\sqrt{v} - 2)/(2v)$, so its minimum occurs at $v = 4$. That minimum is $2 - 2 \log 2 > 0$, since $\log 2 = \int_1^2 t^{-1} dt < 1$. In particular $\log z \leq \sqrt{z}$. Using the exact coefficient $c_{KV} = 55241/1000 < 56$ gives

$$2\nu y = \frac{2y}{c_{KV} z^{2/3} (\log z)^{1/3}} \geq \frac{2y}{c_{KV} z^{5/6}} \geq \frac{2}{c_{KV} 3^{5/6}} y^{1/6} \geq \frac{y^{1/6}}{84}. \quad (56)$$

All quantities raised to fractional powers are positive. For $r = y^{1/6}/84 > 0$, Taylor's theorem with nonnegative remainder gives $e^r \geq r^{18}/18!$. Therefore

$$y^3 e^{-2\nu y} \leq y^3 e^{-r} = 84^{18} r^{18} e^{-r} \leq 18! 84^{18} = C_*.$$

This proves (54) without a numerical evaluation or an unspecified sufficiently-large threshold.

Lemma 10, the envelope $2B(Z_*) \leq 1 - 2\nu$, and $X \geq 1$ yield

$$L(X, T) - 2\pi\Phi(X, T) = O(X^{1-2\nu} \log^3 T) + O(X).$$

Applying (54) proves (53). To retain the separate error accounting, when $T \geq 5$ the cutoff satisfies $Z_* \geq 2T$, and Lemma 10 gives

$$E_{\text{trunc}} = O(X), \quad E_{\text{height}} = O(C_*T), \quad E_{\text{extension}} = O(C_*T), \quad E_{\text{extension}} \leq 0.$$

Here $\log^2 T \leq \log^3 T$ absorbs the extension error, and $\log Z_* \leq 3 \log T$ controls the truncation error. For $3 \leq T < 5$, use the direct $L - 2\pi\Phi = O(X)$ argument already proved in Lemma 10; no assertion $Z_* \geq 2T$ is needed there. Thus $T = 3$, the transition at $T = 5$, and $X = 1, T$ are all covered. The constants are independent of X, T , and all estimates are pointwise at finite parameters. No new sum/integral/limit interchange, RH assumption, simplicity assumption, or substitution for β enters this deduction. $\square$

Equation (53) reconstructs [1, (2.18)], with the locator referring to arXiv:2306.04799v1. The next sections reconstruct the prime-side mean square needed to assemble the pair-correlation asymptotic.

11 A local Fourier mean-value bound

We use $\widehat{f}(\xi) = \int_{\mathbb{R}} f(t) e^{-2\pi i t \xi} dt$ and $e(u) = e^{2\pi i u}$. The bandwidth below is an auxiliary frequency parameter, not a Fourier-support assertion about a zero correlation test function.

Lemma 13 (PAIR-MEANVALUE-001). *Let $\mathcal{M} \subset \mathbb{R}$ be countable and $c_\mu \in \mathbb{C}$ satisfy $\sum_{\mu \in \mathcal{M}} |c_\mu| < \infty$. Set $F(t) = \sum_{\mu} c_\mu e(\mu t)$ and $D = \sum_{\mu} |c_\mu|^2$. Uniformly for $T \geq 1$, $1/(2T) \leq \delta \leq 1/2$,*

$$\int_0^T |F(t)|^2 dt = TD + O \left(\delta^{-1} D + T \sum_{\substack{\mu, \lambda \in \mathcal{M} \\ 0 < |\mu - \lambda| < \delta}} |c_\mu c_\lambda| \right), \quad (57)$$

with an absolute effective constant. The sum is over ordered pairs of different frequencies; repeated frequencies must first be combined.

Proof. Write $s(u) = \sin(\pi u)/(\pi u)$, with $s(0) = 1$, and define

$$K(u) = \frac{3}{4} s(u/2)^4, \quad B_0(u) = s(u/2)^2 + s((u-1)/2)^2, \quad \tau(\xi) = (1 - |\xi|)_+.$$

Here B_0 is unrelated to the zero envelope $B(Z)$. The identity $s(u) = \int_{-1/2}^{1/2} e(u\xi) d\xi$, followed by convolution of the two interval indicators, gives $\widehat{s^2} = \tau$. These identities can be obtained by L^2 Fourier inversion: the interval indicator and s are in L^2 , and $s^2 \in L^1$. The subsequent compactly supported convolutions are absolutely integrable, so Fubini and Fourier inversion give

$$\widehat{K}(\xi) = \frac{3}{2} (\tau * \tau)(2\xi), \quad \widehat{B_0}(\xi) = 2(1 + e(-\xi))\tau(2\xi). \quad (58)$$

For example multiplication of the inverse integral for τ by itself is justified by its finite L^1 norm; its convolution is the transform of the integrable function s^4 . Thus $K, B_0 \geq 0$ are in L^1 ,

$$\int K = \frac{3}{2} \int_{-1}^1 (1 - |v|)^2 dv = 1, \quad \int B_0 = 4,$$

and their Fourier supports are respectively $[-1, 1]$ and $[-1/2, 1/2]$.

Put $H(r) = \int_{|v| \geq r} K(v) dv$, $r \geq 0$. The elementary sinc bounds imply

$$H(r) \leq 1, \quad H(r) \leq \frac{8}{\pi^4 r^3} \quad (r > 0).$$

Moreover, away from the removable points $u = 0, 1$,

$$B_0(u) = \frac{4}{\pi^2} \left(\frac{\sin^2(\pi u/2)}{u^2} + \frac{\cos^2(\pi u/2)}{(u-1)^2} \right) \geq \frac{2}{\pi^2(1+u^2)}.$$

Continuity covers those two points. If $|u| \leq 1$, this lower bound is at least $1/\pi^2$; if $|u| \geq 1$, it is at least $1/(\pi^2 u^2)$. The two bounds for H therefore show

$$H(|u|) \leq 64B_0(u) \quad (u \in \mathbb{R}). \quad (59)$$

The constant 64 is a convenient unoptimized choice.

Let $\chi = 1_{[0, T]}$, $K_\delta(t) = \delta K(\delta t)$, and set

$$M^\pm(t) = (\chi * K_\delta)(t) \pm 64\{B_0(\delta t) + B_0(\delta(t-T))\}.$$

A change of membership in $[0, T]$ under translation by v requires $|v| \geq |t|$ or $|v| \geq |t-T|$. Since $K_\delta \geq 0$ and $\int K_\delta = 1$, this gives, almost everywhere,

$$|\chi * K_\delta(t) - \chi(t)| \leq H(\delta|t|) + H(\delta|t-T|).$$

Hence $M^- \leq \chi \leq M^+$. Endpoints of χ do not affect any integral. Both functions are in L^1 , with

$$\widehat{M}^\pm(0) = T \pm 512/\delta, \quad \|M^\pm\|_1 \leq T + 512/\delta \leq 1025T.$$

Their transforms vanish for $|\xi| > \delta$, by (58), scaling and convolution. Continuity of an L^1 Fourier transform also makes them vanish at $|\xi| = \delta$.

The series for F converges absolutely uniformly on $\mathbb{R}$. In particular Fubini applies to its squared modulus against $M^\pm$, with absolute integral majorant $\|M^\pm\|_1(\sum |c_\mu|)^2$. Therefore

$$\int_{\mathbb{R}} M^\pm(t) |F(t)|^2 dt = \sum_{\mu, \lambda} c_\mu \overline{c_\lambda} \widehat{M}^\pm(\lambda - \mu).$$

The diagonal is $(T \pm 512/\delta)D$; only $0 < |\mu - \lambda| < \delta$ can contribute off the diagonal, each with multiplier at most $1025T$ in modulus. Sandwiching $\int_0^T |F|^2$ between these two integrals proves the claim. The lower function need not be nonnegative: its pointwise lower bound is multiplied by the nonnegative function $|F|^2$. $\square$

12 Prime-pair counts and the diagonal

For $X \geq 1$, put

$$a_X(n) = \frac{\Lambda(n)}{\sqrt{n}} \min\{n/X, X/n\}, \quad D(X) = \sum_{n \geq 1} a_X(n)^2, \quad \ell_X = \log(2X). \quad (60)$$

All integer cutoffs below have full weight. A proper prime power means p^k with $k \geq 2$.

Lemma 14 (PAIR-PRIME-COUNT-001). *Uniformly for real $U \geq 2, V \geq 1$,*

$$\sum_{\substack{1 \leq m, n \leq U \\ 0 < |m-n| \leq V}} \Lambda(m)\Lambda(n) \ll UV. \quad (61)$$

For $0 \leq V < 1$ this sum is empty. Also, uniformly for $X \geq 1, 0 < \delta \leq 1/2$,

$$Q(X, \delta) := \sum_{\substack{m, n \geq 1 \\ 0 < |\log(n/m)| < 2\pi\delta}} a_X(n)a_X(m) \ll \delta X. \quad (62)$$

All pairs are ordered and all constants are absolute and effective.

Proof. First restrict (61) to primes. For even $h > 0$, the sieve input bounds the weighted pairs with $n - m = h$ by $CUg(h)$ when $U \geq 4$, since both logarithmic weights are at most $\log U$. For an odd shift one of the primes is 2, so these pairs contribute $O(V \log^2 U)$. To average g , extend its product definition to all positive integers. Expanding that finite product and then using nonnegativity,

$$\sum_{1 \leq h \leq V} g(h) \leq V \sum_{\substack{d \geq 1 \\ d \text{ odd, squarefree}}} \frac{1}{d} \prod_{p|d} \frac{1}{p-2} = V \prod_{p>2} \left(1 + \frac{1}{p(p-2)}\right) \ll V.$$

The product is bounded by $\exp(\sum_{n \geq 3} 1/(n(n-2))) = \exp(3/4)$. Finite products increase to the displayed series by monotone convergence; the bound proves convergence. Reversing the ordered pairs supplies negative shifts.

There are at most $\sum_{2 \leq k \leq \log_2 U} U^{1/k} \ll \sqrt{U} \log(2U)$ proper prime powers up to U . For each such occurrence at most $2\lceil V \rceil$ other integers lie at positive distance at most V . Bounding both weights by $\log U$, and allowing double counting when both entries are proper powers, gives

$$O(V\sqrt{U} \log^3(2U)) = O(UV).$$

The last inequality is effective because $\log^3(2U)/\sqrt{U}$ is bounded on $U \geq 2$, as follows by putting $w = \log(2U)$ and maximizing $w^3 e^{-w/2}$. Likewise $V \log^2 U = O(UV)$. The finitely many integers involved when $2 \leq U < 4$ are covered directly. This proves (61), including all proper-power pairs.

For the second claim, the close-frequency condition implies

$$e^{-\pi} n < m < e^{\pi} n, \quad |m - n| \leq C_0 \delta n, \quad C_0 = 2\pi e^{\pi},$$

by the mean value theorem for the exponential. If $n \leq X$, use the valid bounds $a_X(n) \leq \Lambda(n)\sqrt{n}/X$ and $a_X(m) \leq \Lambda(m)\sqrt{m}/X$. Their product is at most $C\Lambda(n)\Lambda(m)/X$. Both indices are at most $e^{\pi} X$ and the difference is at most $C_0 \delta X$. Equation (61) bounds this part by $C\delta X$; if the difference cutoff is less than 1, there are no pairs.

For $n > X$, partition into $N < n \leq 2N$, $N = 2^j X$, $j \geq 0$. The bounds $a_X(n) \leq X\Lambda(n)n^{-3/2}$, valid for all n , give a product at most $CX^2\Lambda(n)\Lambda(m)/N^3$. Both indices are at most $2e^{\pi} N$ and their difference is at most $2C_0 \delta N$. Applying the same average, or the empty cutoff case, bounds this block by $C\delta X^2/N$. The nonnegative dyadic sum is bounded by $C\delta X \sum_{j \geq 0} 2^{-j}$, proving (62). $\square$

Lemma 15 (PAIR-PRIME-DIAGONAL-001). *Uniformly for real $X \geq 1$,*

$$D(X) = \log X + O(1), \quad D(X) \ll \ell_X. \quad (63)$$

The proper-prime-power part satisfies

$$D_{\text{pp}}(X) := \sum_{\substack{n=p^k \\ k \geq 2}} a_X(n)^2 \ll X^{-1/2} \ell_X^3 \ll 1. \quad (64)$$

The two portions of the full diagonal are exactly

$$D(X) = X^{-2} \sum_{n \leq X} n \Lambda(n)^2 + X^2 \sum_{n > X} \frac{\Lambda(n)^2}{n^3}.$$

Thus an integer prime power at X is counted once.

Proof. Let $A_p(u) = \sum_{p \leq u} (\log p)^2$. Partial summation of (8) gives

$$A_p(u) = \vartheta(u) \log u - \int_2^u \frac{\vartheta(v)}{v} dv = u \log u + O(u) \quad (u \geq 2).$$

Indeed $(\log u)e^{-c\sqrt{\log u}}$ is bounded effectively, and the integral error is $O(u)$. Enlarging the constant extends this statement to $1 \leq u < 2$. Define the continuous weight

$$w_X(u) = \begin{cases} u/X^2, & 1 \leq u \leq X, \\ X^2/u^3, & u > X. \end{cases}$$

Writing $A_p(u) = u \log u + E_p(u)$, $|E_p(u)| \leq Cu$, Stieltjes integration by parts yields

$$\sum_p w_X(p) (\log p)^2 = \int_1^\infty w_X(u) (\log u + 1) du + O(1) = \log X + 1 - \frac{1}{4X^2} + O(1).$$

Here $E_p(1) = 0$, the boundary term at infinity vanishes, and

$$\int_1^\infty u |w'_X(u)| du = X^{-2} \int_1^X u du + 3X^2 \int_X^\infty u^{-3} du \leq 2.$$

The weight agrees from both sides at X , so there is no jump term, even when X is prime. All improper integrals converge absolutely; the argument is first on finite intervals, then takes the upper endpoint to infinity at fixed X .

Counting all integer k -th powers, as in the elementary estimate $\sum_{2 \leq k \leq \log_2 u} u^{1/k} \ll \sqrt{u} \log(2u)$, gives

$$A_{\text{pp}}(u) := \sum_{\substack{p^k \leq u \\ k \geq 2}} (\log p)^2 \ll \sqrt{u} \log^3(2u).$$

For $n \leq X$, use $w_X(n) \leq X^{-1}$. For $2^j X < n \leq 2^{j+1} X$, use $w_X(n) \leq 2^{-3j}/X$. Consequently

$$D_{\text{pp}}(X) \ll X^{-1/2} \ell_X^3 + X^{-1/2} \sum_{j \geq 0} 2^{-5j/2} (\ell_X + (j+1) \log 2)^3 \ll X^{-1/2} \ell_X^3.$$

The series is a convergent nonnegative majorant with an absolute effective constant, since $\ell_X \geq \log 2$. Maximizing a polynomial times an exponential as above shows the last expression is bounded on $X \geq 1$. Adding primes and proper powers proves the assertions. $\square$

13 The weighted prime-series mean square

Lemma 16 (PAIR-PRIME-MEAN-001). *Uniformly for $T \geq 3$ and $1 \leq X \leq T$,*

$$\int_0^T |\mathcal{P}(X, t)|^2 dt = TD(X) + E_{\text{offdiag}}(X, T), \quad |E_{\text{offdiag}}| \ll \sqrt{TX\ell_X}, \quad (65)$$

$$TD(X) = T \log X + E_{\text{diagonal}}(X, T), \quad |E_{\text{diagonal}}| \ll T. \quad (66)$$

In particular $\|\mathcal{P}(X, \cdot)\|_{L^2(0, T)} \ll \sqrt{T\ell_X}$. All constants are absolute and effective.

Proof. Take frequencies $\mu_n = -\log n/(2\pi)$, $n \geq 2$, and real coefficients $a_X(n)$. For fixed X , the coefficient sum is finite: the tail is bounded by $X \sum_{n>X} (\log n) n^{-3/2}$. More explicitly, for $N \geq \max\{2, X\}$,

$$\sup_{t \in \mathbb{R}} \left| \sum_{n>N} a_X(n) n^{-it} \right| \ll X \frac{\log N + 1}{\sqrt{N}}.$$

This also justifies finite-sum approximation of the squared integral at fixed X, T . No uniformity in unbounded X is required to take this limit.

Choose

$$\delta = \frac{1}{2} \sqrt{\frac{\ell_X}{TX}}. \quad (67)$$

For $X \geq 1$, $\ell_X \leq X$, since $X - \log(2X)$ is increasing and positive there. Also ℓ_X/X increases up to $X = e/2$ and then decreases; its minimum on $[1, T]$ is at an endpoint. Thus

$$T\ell_X/X \geq \min\{T \log 2, \log(2T)\} > 1 \quad (T \geq 3).$$

For example $\log 2 > 1/2$ follows by integrating $1/u$ on $[1, 2]$, and $\log(2T) > \log 3 > 1$. It follows that $1/(2T) \leq \delta \leq 1/2$, including both endpoints $X = 1, T$.

Apply Lemma 13, translating $|\mu_m - \mu_n| < \delta$ exactly into $|\log(m/n)| < 2\pi\delta$. Lemmas 14 and 15 give

$$|E_{\text{offdiag}}| \ll \delta^{-1} D(X) + TQ(X, \delta) \ll \delta^{-1} \ell_X + T\delta X \ll \sqrt{TX\ell_X}.$$

This error includes the diagonal effect of the Fourier majorants; its name denotes the total error after subtracting $TD(X)$. The diagonal lemma then gives (66). Finally $X \leq T$ and $\ell_X \geq \log 2$ imply $\sqrt{TX\ell_X} \ll T\ell_X$, proving the norm bound. $\square$

14 The full prime-side squared norm

For $t \geq 0$, define exact functions

$$\begin{aligned} A(X, t) &= X^{-1} \log(t+2), & H(X, t) &= E_{\text{arch}} + E_{\text{DS}} + E_{\text{pole}} + E_{\text{trivial}}, \\ r(X, t) &= -\mathcal{P}(X, t) + A(X, t) + H(X, t). \end{aligned}$$

where the four remainders are those in Lemma 6. Set $R(X, T) = \int_0^T |r(X, t)|^2 dt$. The exact explicit formula gives $r = \mathcal{S}$, so $R = L$ with the existing normalization. We evaluate R from its prime-side expression.

Lemma 17 (PAIR-RHS-MEAN-001). *Uniformly for $T \geq 3$ and $1 \leq X \leq T$,*

$$R(X, T) = \frac{T}{X^2} \log^2 T + T \log X + O\left(\frac{T \log T}{X^2}\right) + O(T\sqrt{\log T}), \quad (68)$$

with absolute effective implied constants and both endpoints included. This is the prime-side form of [1, (2.19)].

Proof. Write $q = \log T$, $\ell = \ell_X$, and use $L^2(0, T)$ norms. The archimedean square has the exact antiderivative

$$\int_0^T \log^2(t+2) dt = [(t+2)\{\log^2(t+2) - 2\log(t+2) + 2\}]_0^T.$$

Since $\log(T+2) - \log T = O(1/T)$, it follows that

$$\|A\|_2^2 = \frac{Tq^2}{X^2} + E_{\text{archsquare}}, \quad E_{\text{archsquare}} = O(Tq/X^2), \quad \|A\|_2 \ll \sqrt{T}q/X. \quad (69)$$

The prime/archimedean interaction is treated by oscillation. Integration by parts, for $n \geq 2$, gives

$$\left| \int_0^T \log(t+2)n^{-it} dt \right| \leq \frac{2\log(T+2)}{\log n}.$$

Also $\Lambda(n)/\log n \leq 1$ and hence

$$\sum_{n \geq 2} \frac{a_X(n)}{\log n} \leq X^{-1} \sum_{2 \leq n \leq X} \sqrt{n} + X \sum_{n > X, n \geq 2} n^{-3/2} \ll \sqrt{X}.$$

The coefficient series is absolutely summable, so its integral against the bounded function $\log(t+2)$ can be interchanged. Thus

$$E_{\text{primearch}} := -2\Re \int_0^T \mathcal{P}(X, t) \overline{A(X, t)} dt \ll \frac{\log(T+2)}{\sqrt{X}} \quad (70)$$

in absolute value.

Keep the four original errors separate initially. Lemma 6 gives

$$\begin{aligned} b_{\text{arch}} &:= \|E_{\text{arch}}\|_2 \ll \sqrt{T}/X, & b_{\text{DS}} &:= \|E_{\text{DS}}\|_2 \ll \sqrt{T}/X, \\ b_{\text{pole}} &:= \|E_{\text{pole}}\|_2 \ll \sqrt{X}, & b_{\text{trivial}} &:= \|E_{\text{trivial}}\|_2 \ll X^{-5/2}. \end{aligned}$$

For the last two bounds use respectively $\int_0^\infty (1+t^2)^{-2} dt < \infty$ and $\int_0^\infty (t+2)^{-2} dt = 1/2$. If j runs over these four errors, their exact contributions to the square, beyond $\|-\mathcal{P} + A\|_2^2$, are

$$E_{P,j} = -2\Re \int_0^T \mathcal{P} \overline{E_j} dt, \quad E_{A,j} = 2\Re \int_0^T A \overline{E_j} dt, \quad E_{\text{remsquare}} = \|H\|_2^2.$$

Cauchy-Schwarz gives $|E_{P,j}| \leq 2\|\mathcal{P}\|_2 b_j$, $|E_{A,j}| \leq 2\|A\|_2 b_j$, and $E_{\text{remsquare}} \leq (\sum_j b_j)^2$. Using Lemma 16 and (69), the separate uniform bounds are as follows; entries bound absolute values.

j	$E_{P,j}$	$E_{A,j}$
arch, DS	$O(T\sqrt{\ell}/X)$	$O(Tq/X^2)$
pole	$O(\sqrt{TX}\ell)$	$O(\sqrt{T/X}q)$
trivial	$O(\sqrt{T}\ell X^{-5/2})$	$O(\sqrt{T}q X^{-7/2})$

The sum of the four b_j is $O(\sqrt{T}/X + \sqrt{X})$, since $T \geq 3, X \geq 1$; therefore

$$0 \leq E_{\text{remsquare}} \ll T/X^2 + X.$$

Expanding the finite sum of five exact functions now gives

$$R = \frac{Tq^2}{X^2} + T \log X + E_{\text{diagonal}} + E_{\text{offdiag}} + E_{\text{archsquare}} + E_{\text{primearch}} + \sum_j (E_{P,j} + E_{A,j}) + E_{\text{remsquare}}.$$

Every term is integrable, by the preceding bounds. To combine errors, use $q > 1$, $X \leq T$, $\ell \leq \log(2T) \ll q$, $\sqrt{\ell}/X \leq 1$, and $q \leq T$. The diagonal is $O(T)$; the off-diagonal and pole/prime interaction are $O(T\sqrt{q})$. The archimedean square and its interactions with arch and DS errors are $O(Tq/X^2)$. The remaining table entries and $E_{\text{primearch}}$, $E_{\text{remsquare}}$ are $O(T\sqrt{q})$. This proves (68) with the two displayed error scales.

All arguments hold on the closed range $T \geq 3, 1 \leq X \leq T$. Prime powers at X are retained once, and $n = 1$ has zero coefficient. The prime-series limits are taken at fixed X, T under absolute majorants; the resulting estimates are uniform in the stated range. No RH, simplicity, zero-density estimate, or prime-pair asymptotic is used. $\square$

15 The normalized pair-correlation asymptotic

Theorem 18 (PAIR-ASYMPTOTIC-001). *Assumptions: UNCONDITIONAL. Let I_T index all nontrivial zero occurrences $\rho_j = \beta_j + i\gamma_j$ with $0 < \gamma_j \leq T$, counted with multiplicity and full weight at T . Set*

$$C_T = \frac{T \log T}{2\pi}, \quad \mathcal{F}_T(\alpha) = \frac{1}{C_T} \sum_{j,k \in I_T} \frac{4T^{\alpha(\rho_j - \rho_k)}}{4 - (\rho_j - \rho_k)^2}.$$

The sum includes all ordered pairs, including equal indices and equal complex zeros. For every $T \geq 3$, this function is real, nonnegative and even on $\mathbb{R}$. There are absolute effective constants $C_1, C_2 > 0$ such that, for every $T \geq 3$ and $|\alpha| \leq 1$,

$$\left| \mathcal{F}_T(\alpha) - T^{-2|\alpha|} \log T - |\alpha| \right| \leq C_1 T^{-2|\alpha|} + C_2 (\log T)^{-1/2}. \quad (71)$$

Equivalently, uniformly on this closed range,

$$\mathcal{F}_T(\alpha) = T^{-2|\alpha|} (\log T + O(1)) + |\alpha| + O((\log T)^{-1/2}). \quad (72)$$

This reconstructs [1, Theorem 1, (1.4)] in the project normalization.

Proof. Reality, nonnegativity and evenness for every real α are Corollary 9. To prove the estimate, first fix $T \geq 3$ and $1 \leq X \leq T$, and put $q = \log T > 1$. The exact explicit formula gives $R = L$. Lemmas 17 and 12 therefore give

$$2\pi\Phi(X, T) = TX^{-2}q^2 + T \log X + E_{\text{peak}} + E_{\text{mean}} + E_{\text{compare}, T} + E_{\text{compare}, X}. \quad (73)$$

Here the first two errors collect the two already justified prime-side error scales; the last two are the errors in $2\pi\Phi - L$. Their signs are immaterial for the following absolute bounds. All implied constants are absolute and effective, uniformly on the closed range just stated:

Error	Before normalization	After division by Tq
E_{peak}	$O(Tq/X^2)$	$O(X^{-2})$
E_{mean}	$O(T\sqrt{q})$	$O(q^{-1/2})$
$E_{\text{compare}, T}$	$O(T)$	$O(q^{-1})$
$E_{\text{compare}, X}$	$O(X)$	$O(X/(Tq))$

In particular, division of (73) gives

$$\begin{aligned} \frac{2\pi\Phi(X, T)}{Tq} &= X^{-2}q + \frac{\log X}{q} + O(X^{-2}) + O(q^{-1/2}) \\ &\quad + O(q^{-1}) + O(X/(Tq)). \end{aligned} \quad (74)$$

Since $X/(Tq) \leq q^{-1} \leq q^{-1/2}$, the last two errors can now be absorbed into $O(q^{-1/2})$. For $0 \leq \alpha \leq 1$, set $X = T^\alpha$. Then $1 \leq X \leq T$, $\log X/q = \alpha$, $X^{-2} = T^{-2\alpha}$, and the left side of (74) is exactly $\mathcal{F}_T(\alpha)$. The uniform absolute error bounds give (71) on this half interval. For $-1 \leq \alpha < 0$, use $\mathcal{F}_T(\alpha) = \mathcal{F}_T(|\alpha|)$. Thus no prime-side estimate is applied with $X < 1$.

This proof introduces no sum, integral, or limit interchange. The identities are used at fixed T, X , followed by a substitution into bounds uniform on the whole closed range. In particular, the asymptotic holds for every choice $\alpha = \alpha(T) \in [-1, 1]$ as $T \rightarrow \infty$. Effectivity follows from the two input estimates; no numerical values for the complete constants C_1, C_2 are claimed. $\square$

At the endpoints the same theorem gives

$$\mathcal{F}_T(0) = \log T + O(1), \quad \mathcal{F}_T(\pm 1) = 1 + T^{-2} \log T + O(T^{-2}) + O((\log T)^{-1/2}) = 1 + O((\log T)^{-1/2}).$$

For the last absorption, writing $q = \log T \geq 1$, the elementary bound $e^{2q} \geq (2q)^2/2 \geq q^{3/2}$ gives $T^{-2}q \leq q^{-1/2}$. At $\alpha = 0$, the separate $O(T^{-2|\alpha|})$ term must be retained; the proof does not yield an $O((\log T)^{-1/2})$ total error there. The normalization remains C_T , not TL_T or $N(T)$. No frequency range beyond $[-1, 1]$, purely vertical unweighted correlation theorem, or horizontal proportion is asserted here.

16 Checks and scope of the audit

The zero-side denominator is at least $3/4$ in modulus when $t = \gamma$; no exceptional real t is excluded. Conjugation of the full zero sum and reindexing $\rho \mapsto \bar{\rho}$ give $\mathcal{S}(X, -t) = \overline{\mathcal{S}(X, t)}$, and every exact term on the right has the same symmetry. The reflection $\rho \mapsto 1 - \bar{\rho}$ preserves multiplicity and the strip bound used in all zero estimates; it reindexes the sum with $\delta_\rho \mapsto -\delta_\rho$, and does not assert equality of individual summands. As a separate check under assumption RH, the zero side reduces to $\sum_\gamma 2X^{i(\gamma-t)}/(1 + (t - \gamma)^2)$, with multiplicity. This specialization is not used in an unconditional proof.

The rational kernel is exactly the subtraction in the source's argument: for $r = \rho - s$,

$$\frac{1}{1-r} - \frac{1}{-1-r} = \frac{2}{1-r^2}.$$

This proof therefore establishes the combined identity directly, without assigning absolute convergence to either separate Landau zero series. There is no Fourier transform or mean-spacing rescaling in this lemma.

For the squared norm, complex conjugation of the integrable pair product exchanges j, k . Its explicit value in (35)–(36) has the same property, since h is unchanged and $d \mapsto -d$. Under the separate check assumption RH, all $\delta_j = 0$, so the line shift disappears and the pair integral becomes $2\pi/(4 + (\gamma_j - \gamma_k)^2)$ times $X^{i(\gamma_j - \gamma_k)}$. The full identity then recovers the usual ordinate kernel with coefficient 4. The formulas also apply directly at $X = 1$ and for $0 < X < 1$, and when I_T is empty both sides vanish.

For height truncation, the bounds hold at $X = 1$, even when the finite zero set is empty or zeros lie exactly on either cutoff. Under the separate check assumption RH, $B(Z) = 0$, so the height and extension bounds reduce to $O(\log^3 T)$ and $O(\log^2 T)$; the distant-zero estimate remains $O(X)$ after the stated choice of Z . This check does not assert a sharp RH-dependent refinement of that last error.

All local statements have status **proved-draft**. The imported analytic inputs are identified above and the local sum/integral exchanges are justified within the proof. The local Lemmas 1–4

reconstruction uses no numerical result. The finite-height envelope PAIR-ENVELOPE-001 and the uniform comparison PAIR-COMPARE-001 additionally use the published inputs ZETA-KV-001 and ZETA-LOW-001, with their external computational dependencies. No local numerical certificate or independent replay is claimed for those inputs. The local RH-contamination review is recorded in the research audit files. Independent mathematical review, upstream proof reconstruction, final journal comparisons and clean-checkout reproduction remain pending. The normalized theorem is reconstructed as PAIR-ASYMPTOTIC-001; it inherits the external computational dependencies of PAIR-COMPARE-001. The prime-side mean square is reconstructed as PAIR-RHS-MEAN-001 from the explicit formula, a local Fourier argument, and the imported unconditional PNT and sieve bounds; this part uses no numerical input. Finite arithmetic regression scripts check the elementary kernel, endpoint, residue, reindexing and truncation algebra, as well as the final main-term and error normalization; they do not certify any analytic estimate.

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